Temporal Leakage in Windowed Time-Series Classification: Systematic Evaluation Bias and a Standardized Splitting Protocol for Fault Diagnosis

Abstract: High accuracy in deep learning-based continuous time-series analysis is often regarded as evidence of robust detection and diagnostic capability. However, this assumption is valid only when the test data are strictly blind and free from data leakage. Using vibration-based fault diagnosis as a representative case, this study systematically investigates how random splitting of overlapping sliding windows introduces temporal data leakage and consequently inflates apparent model performance. A comprehensive evaluation assesses four mainstream architectures on a self-built eccentric gear dataset and the CWRU dataset. Experimental results indicate that under random splitting with a 75% overlap rate, the ResNet-1D model achieves an accuracy of 1.00±0.00. However, after enforcing strict temporal isolation, its performance plummets to near random-guessing levels. Models tend to exploit redundant waveform segments as shortcut representations rather than learning physically meaningful fault features. To address this evaluation limitation, a Standardized Data Splitting Protocol (SDSP) is proposed. By enforcing block-level temporal isolation and a safety margin (ΔT) while retaining intra-set overlap for data augmentation, SDSP explicitly decouples evaluation integrity from sample expansion at the operational level. Furthermore, a Leakage Impact Factor (*LIF*) and a Generalization Deception Gap (ΔGap) are introduced. These serve as supplementary metrics to quantify model vulnerability to temporal redundancy. This research demonstrates that a single accuracy metric is insufficient to reliably evaluate the true performance of windowed time series learning. It also provides a rigorous framework for evaluating trustworthy deep learning in continuous monitoring applications.

Keywords: Fault diagnosis; Data leakage; Sliding window; Time series evaluation; Deep learning; Standardized Data Splitting Protocol (SDSP)

1. Introduction

Deep learning has become the mainstream method for continuous time-domain signal analysis in fields such as industrial condition monitoring, biomedical sensing, energy systems, and autonomous perception [1-4]. In vibration signal-based fault diagnosis, raw sensor data streams are typically segmented into fixed-length sliding window samples for model training and evaluation [5]. This routine operation generates numerous reports of high-accuracy or even near-perfect test results [6-9]. However, high accuracy holds practical significance only when the evaluation protocol provides a truly independent test set [10].

Continuous time series data exhibits strong temporal and physical dependencies between adjacent segments [11-12]. Sliding window sampling further amplifies this dependency. Overlapping windows directly share waveform segments. Even non-overlapping adjacent windows often remain significantly correlated due to physical mechanisms like mechanical inertia, low-frequency background vibration, and slowly varying operating conditions. When these window samples are randomly split into training and test sets, temporally adjacent information flows across subset boundaries. Consequently, the test set is not a true blind set. The extremely high accuracy exhibited by models likely reflects mere memorization of local repetitive waveform patterns rather than true generalization ability to unseen conditions.

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This issue profoundly impacts both engineering deployment and theoretical research. Deep learning models are inherently highly nonlinear black-box systems [13-15]. Although explainable artificial intelligence is introduced into fault diagnosis to reveal model decision bases [16-18], the validity of such explanations fundamentally depends on the authenticity of the learned representations. If models train under temporal leakage conditions, their "decision bases" rely on fitting local waveform redundancy rather than genuinely encoding physical fault mechanisms. This leads to misleading conclusions in any interpretability analysis. More severely, numerous current studies use test accuracy under random splitting as a key criterion for evaluating algorithms. If the data splitting method itself possesses fundamental flaws, accuracy comparisons built upon it lose credibility. The claimed performance advantages of related algorithms also require re-examination. As the most upstream design decision in the supervised learning pipeline, the rationality of data splitting directly determines the validity boundary of all subsequent performance claims. Only by establishing scientific and rigorous data splitting and evaluation protocols is it possible to ensure the reliability of developed algorithms under truly unknown conditions. This ultimately drives deep learning-based fault diagnosis technologies from the laboratory to industrial application.

To address the aforementioned evaluation limitations, this paper conducts a systematic study. Experiments primarily target an eccentric gear fault diagnosis platform under multiple drivetrain configurations and rotational speeds. Cross-dataset validation is also performed on the CWRU bearing benchmark dataset. The evaluation covers four representative architectures: WDCNN, ResNet-1D, C-BiLSTM, and Transformer. The core focus here is not to explore which architecture achieves the highest accuracy. Instead, it investigates the true generalization performance boundaries of models under strict conditions that completely eliminate temporal redundancy between subsets. This aims to reshape a more scientific and rigorous evaluation benchmark for the field. To this end, a Standardized Data Splitting Protocol (SDSP) is proposed. SDSP eliminates evaluation leakage through the joint constraints of block-level temporal isolation and a temporal safety margin (ΔT). Furthermore, a Leakage Impact Factor (*LIF*) and a Generalization Deception Gap (ΔGap) are introduced as supplementary metrics for quantitative diagnosis of the leakage extent.

The main contributions of this paper are as follows:

1. **Revealing evaluation bias and performance inflation mechanisms**: Systematic empirical analysis demonstrates that random overlapping splitting introduces evaluation leakage across multiple architectures, datasets, and operating conditions. This suggests that the high accuracy exhibited by existing models likely stems from reasoning and fitting based on contextual temporal cues, rather than true recognition of underlying physical fault features. This provides a new perspective for understanding actual model decision mechanisms.
2. **Proposing the Standardized Data Splitting Protocol (SDSP)**: By introducing block-level splitting and a configurable safety margin (ΔT) to enforce temporal isolation, SDSP explicitly decouples data augmentation (allowing intra-set overlap) from evaluation integrity (enforcing inter-set isolation) at the operational level.
3. **Constructing quantitative leakage diagnostic metrics**: The Leakage Impact Factor (*LIF*) and Generalization Deception Gap (ΔGap) are introduced. These provide novel quantitative evaluation tools for assessing the reliability of model performance under

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traditional evaluation protocols.

4. **Providing mechanistic evidence of shortcut learning**: Feature space visualization and training dynamic analysis reveal the internal mechanisms by which temporal leakage induces shortcut learning. This also highlights the potential failure risks of interpretability analysis under leakage conditions.
5. **Cross-platform validation of evaluation bias universality**: Cross-dataset validation confirms that this evaluation bias is independent of specific experimental platforms. This further establishes the necessity of leakage-free evaluation protocols in continuous monitoring applications.

The remainder of this paper is organized as follows. Section 2 reviews relevant research and highlights the limitations of existing evaluation paradigms. Section 3 elaborates on the SDSP framework and leakage quantification methods. Section 4 conducts systematic experimental validation under multi-condition and multi-architecture settings, including cross-platform validation using the CWRU dataset. Section 5 provides an in-depth discussion of the findings. Finally, Section 6 summarizes the paper and outlines future research directions.

2. Related Work

2.1 Deep Learning Applications in Fault Diagnosis

Over the past decade, intelligent fault diagnosis experiences a fundamental shift from manual feature engineering to end-to-end data-driven learning [19,20]. Early methods rely on manual features such as time-domain statistics, frequency-domain amplitude spectra, and time-frequency distributions. These combine with support vector machines or shallow neural networks to achieve fault classification [21-26]. Such methods highly depend on domain prior knowledge and show limited generalization under varying operating conditions. Deep convolutional neural networks, represented by WDCNN, prove that end-to-end learning from raw vibration waveforms rivals or even surpasses feature engineering pipelines [27,28]. This triggers widespread CNN applications in bearing, gear, and rotor fault diagnosis [29-31]. ResNet-1D solves the vanishing gradient problem in deep networks via cross-layer identity mapping [32,33]. LSTM and BiLSTM explicitly model sequence dependencies in vibration signals [34-36]. Furthermore, the hybrid architecture C-BiLSTM combines convolutional feature extraction with bidirectional recurrent modeling. This further enhances representation capability for non-stationary signals [37]. The Transformer architecture, based on multi-head self-attention mechanisms, shows powerful performance in long-range temporal modeling tasks. It sees wide application in both single-domain and cross-domain fault diagnosis [38-40].

Although these architectures achieve remarkable test accuracy on standard benchmark datasets, the evaluation methods in most published results remain largely unchanged. Models train and test on window samples randomly split from the same continuous signal. While this practice facilitates direct comparison between architectures, its risk as a potential source of systematic evaluation bias remains insufficiently explored. When the evaluation protocol itself contains temporal leakage, accuracy differences between architectures partially stem from their varying abilities to fit local waveform redundancy. They do not necessarily reflect true discriminative capabilities regarding underlying physical fault features.

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2.2 Transfer Learning and Domain Adaptation

Given the practical constraints of scarce labeled data and variable operating conditions in real industrial scenarios, transfer learning and domain adaptation methods receive widespread attention in fault diagnosis [41-48]. These methods attempt to transfer feature representations learned in the source domain to the target domain through domain alignment, adversarial training, meta-learning, or instance reweighting. This compensates for performance degradation across machines, speeds, or load conditions [49-53]. Numerous studies habitually attribute the accuracy gap between source and target domains to domain shift. They subsequently use this as the core motivation for designing domain adaptation algorithms.

However, this attribution implies an insufficiently verified premise: the source domain model truly learns physically meaningful and transferable fault features during the training phase. If the source domain model trains under temporal leakage conditions, its learned representations rely more on memorizing local waveform redundancy. It does not genuinely extract transferable physical fault features. Such representations perform well in random splitting evaluations within the source domain but experience significant performance degradation during domain transfer. This degradation partially arises because the model fails to learn truly cross-domain universal physical features, rather than from the domain shift itself. From this perspective, the practical significance of the "transfer performance improvement" observed in domain adaptation methods requires further validation under a leakage-free evaluation framework. In other words, before exploring how to overcome domain shift, it is necessary to ensure that the model truly learns transferable physical features in the source domain. Therefore, constructing a rigorous, leakage-free evaluation protocol is not only necessary to verify the true performance of existing algorithms. It also serves as a key cornerstone for driving substantial breakthroughs in domain adaptation technologies.

2.3 Physics-Informed Fusion and Interpretable Diagnosis

The limitations of purely data-driven methods in small-sample and high-noise scenarios drive the rise of physics-informed deep learning in fault diagnosis [54-60]. Typical implementation paths include using fault characteristic frequencies as initialization constraints for learnable filters [54], introducing physical consistency regularization terms into the loss function [55], and generating simulation samples based on dynamic models to expand the training set [56,57]. Meanwhile, introducing attention mechanisms and residual shrinkage networks further enhances the adaptive focusing ability of models on fault-sensitive features [61,62]. Visualizations of attention weights and soft threshold parameters widely support claims of model interpretability.

However, the claimed performance gains and interpretability of these two types of methods depend to some extent on the same premise: the learned representations truly reflect the physical mechanisms of faults. If the model trains under temporal leakage conditions, representation improvements brought by physical constraints mix with leakage-induced performance inflation. This makes it difficult to accurately isolate the true source of algorithm performance gains. The "focus patterns" presented by attention weights also partially reflect the memorization of repetitive waveform segments in the training set, rather than true perception of physical fault features. This poses potential challenges to the reliability of related interpretability conclusions.

In summary, existing research makes significant progress in architectural innovation, transfer learning, physics-informed fusion, and interpretability methods. However, the temporal rationality

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of data splitting protocols lacks systematic review and standardization. This rationality is the core premise ensuring a credible foundation for the aforementioned research conclusions. Based on this, the present study attempts to systematically examine and standardize this long-neglected evaluation blind spot. By establishing a standardized data splitting protocol, it expects to provide a more solid and reliable benchmark for the future development and comparison of fault diagnosis algorithms.

3. Methodology

3.1 Based on Information Redundancy Quantification for Temporal Decoupling

In the framework of data-driven fault diagnosis, the independent and identically distributed assumption of samples is the statistical prerequisite for evaluating model generalization. Let the original continuous vibration sequence be denoted as X. A sample set D is extracted using a sliding window operator. Specifically, the generation process of the i-th sample x_i can be mathematically formulated as:

$$x_i = \{X[t]\}_{t=i\sigma}^{i\sigma+\omega-1} \tag{2}$$

where ω denotes the window length and σ represents the sliding step size. The overlap rate $\gamma \in [0,1)$ between adjacent samples is defined as:

$$\gamma = 1 - \frac{\sigma}{\omega} \tag{3}$$

During model construction, the complete sample set D must be partitioned into a training set D_{tr}, a validation set D_{va}, and a testing set D_{te}. Here, D_{tr} is utilized to minimize empirical risk, D_{va} is used for hyperparameter tuning and early stopping, and D_{te} serves as an absolute blind test set for final performance measurement. To standardize the data scale under different evaluation strategies, we define a global split ratio vector $\Theta=[\alpha,\beta,\delta]$, where α, β, and δ represent the proportions of each subset relative to the total sample size N, strictly satisfying the following capacity conservation constraint [23]:

$$\alpha + \beta + \delta = 1, \quad \{\alpha, \beta, \delta\} \in (0,1) \tag{4}$$

From an information-theoretic perspective, when $\gamma>0$, adjacent samples x_i and x_{i+1} share a common sequence of length $\gamma\omega$. This overlap results in non-zero mutual information $I(D_{tr}; D_{te})>0$ between the training and testing sets. However, satisfying a zero-overlap constraint ($\gamma=0$) alone is insufficient to completely block temporal leakage caused by low-frequency background noise and mechanical inertia. Even without sample overlap, temporally adjacent samples still exhibit extremely high dynamical similarity. Therefore, the Standardized Data Splitting Protocol (SDSP) introduces a Temporal Safety Gap ΔT. Defining the temporal boundary of the training set as T_{tr}^{end} and the starting time of the testing set as T_{te}^{start}, the protocol mandates:

$$T_{te}^{start} - T_{tr}^{end} \geq \Delta T \tag{5}$$

where ΔT is typically set to a multiple of the window length (e.g., $\Delta T=k\cdot\omega$) to ensure true physical decoupling of dynamical states across subsets. Consequently, SDSP enforces an asymmetric overlap constraint: it permits $\gamma>0$ within the training set for reasonable data augmentation, but strictly requires zero overlap and a mandatory safety gap ΔT between D_{tr} and D_{te}, thereby

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eradicating evaluation biases induced by local waveform similarities.

Fig. 1 illustrates the fundamental differences among three representative data splitting strategies. Random Split distributes overlapping windows arbitrarily across subsets, introducing severe temporal leakage throughout the entire dataset. Sequential Split preserves temporal order but still suffers from boundary leakage at the transition zones between subsets. In contrast, the proposed Block-Holdout Split under the SDSP protocol enforces explicit safety gaps ΔT at both boundaries, ensuring complete temporal isolation between the training, validation, and testing sets.

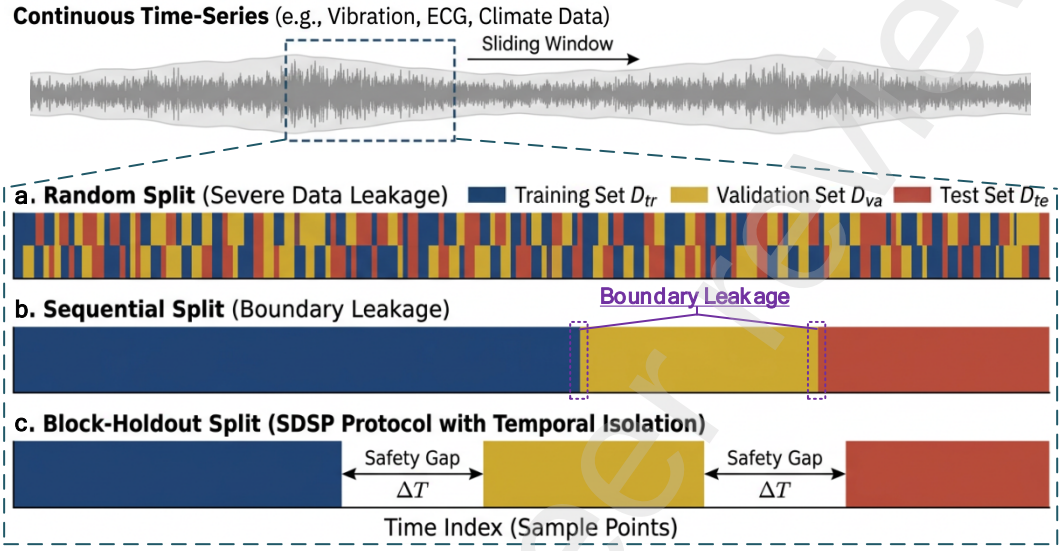


Fig. 1. Comparison of three data splitting strategies for continuous time-series.

3.2 Quantification of Data Leakage Sensitivity

To objectively evaluate the resistance of different feature extraction operators (e.g., CNN, RNN, Transformer) to data leakage and their propensity for Shortcut Learning, the SDSP framework defines the Leakage Impact Factor (*LIF*). Assuming a model achieves a testing accuracy of Acc_{Random} under traditional random splitting (which suffers from severe temporal leakage) and a true accuracy of Acc_{SDSP} under strict SDSP compliance (i.e., block-holdout isolation with temporal decoupling and safety gap ΔT), *LIF* is defined as:

$$LIF = \frac{Acc_{Random} - Acc_{SDSP}}{Acc_{Random}} \times 100\% \tag{6}$$

A higher *LIF* indicates that the algorithm's superior performance relies heavily on memorizing overlapping redundant information, rendering its generalization capability highly unreliable in real-world industrial scenarios.

Furthermore, traditional random splitting often masks true overfitting, creating a false illusion of high consistency between training and testing performance. To address this, the protocol introduces the ΔGap:

$$\Delta Gap = \left| (Acc_{tr} - Acc_{te})_{SDSP} - (Acc_{tr} - Acc_{te})_{Random} \right| \tag{7}$$

where Acc_{tr} and Acc_{te} represent the training and testing accuracies under the corresponding splitting strategy. ΔGap quantitatively characterizes the extent to which traditional evaluation methods conceal the model's true generalization gap. Through the joint constraint of *LIF* and ΔGap, the SDSP protocol provides a standardized mathematical criterion for the industry to screen

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robust algorithms capable of capturing genuine physical features.

3.3 Feature Space Clustering Evaluation

To quantify the quality of feature representations learned by the models, we employ the Silhouette Score. For any sample i in the feature space belonging to class C_I, its intra-class average distance $a(i)$ is defined as the mean Euclidean distance to all other samples within the same class [25]:

$$a(i) = \frac{1}{|C_I| - 1} \sum_{j \in C_I, i \neq j} d(i, j) \tag{8}$$

where $|C_I|$ is the total number of samples in class C_I, and $d(i,j)$ denotes the Euclidean distance between samples i and j in the high-dimensional feature space. A smaller $a(i)$ indicates that the sample is more compactly clustered within its class. Concurrently, the minimum inter-class average distance $b(i)$ for sample i is defined as the minimum of the mean distances to all samples in any other class C_J ($J \neq I$):

$$b(i) = \min_{J \neq I} \left(\frac{1}{|C_J|} \sum_{j \in C_J} d(i, j) \right) \tag{9}$$

A larger $b(i)$ indicates that sample i is well-separated from other classes. Based on these definitions, the silhouette coefficient $s(i)$ for a single sample i is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max\left(a(i), b(i)\right)} \tag{10}$$

The global Silhouette Score S for the entire dataset is the arithmetic mean of the silhouette coefficients of all N samples:

$$S = \frac{1}{N} \sum_{i=1}^{N} s(i) \tag{11}$$

The value of S ranges from $[-1,1]$. When S approaches 1, it indicates extremely high clustering quality in the feature space; when S approaches 0, it signifies severe overlapping of feature boundaries between different classes; a negative S indicates that the feature space is completely disordered.

4. Experimental Validation

To systematically evaluate the impact of data leakage induced by overlapping sampling on vibration signal fault diagnosis, experimental validation is conducted from four dimensions based on an eccentric gear transmission test platform and the CWRU bearing benchmark dataset. As shown in Fig. 2(a), the eccentric gear test rig consists of five core components: a drive motor, a test gearbox, a coupling, a load gearbox, and a load motor. Based on the transmission chain length, the setup is divided into four physical configurations (T1, T2, T3, and T4, with the eccentric gear positions marked in red in Fig. 2) to simulate fault transmission paths at different spatial locations in industrial settings. Furthermore, the CWRU bearing dataset serves as an external validation benchmark to verify the universality of the discovered patterns (Fig. 2(b)).

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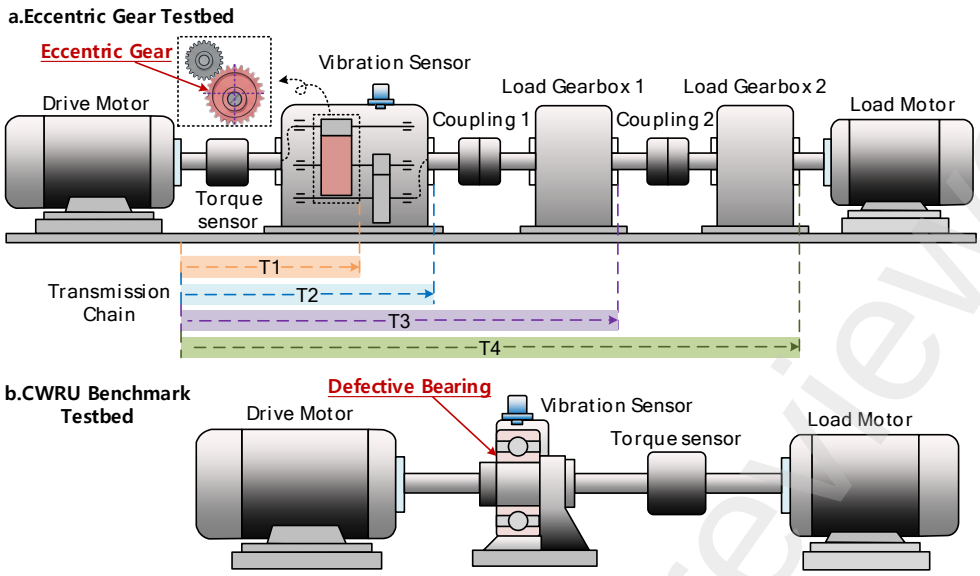
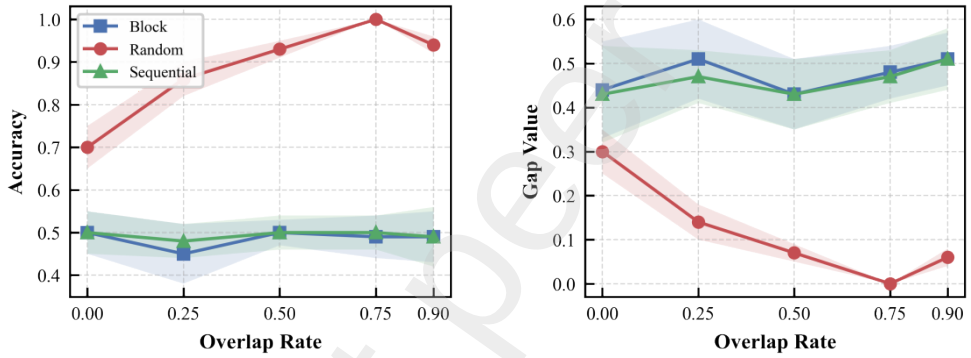


Fig. 2. Gear eccentricity and CWRU fault diagnosis test rigs.



(a) (b)

Fig. 3. Data leakage mechanisms under different splitting strategies for T3 s600. The evaluation is based on a ResNet-1D model with a standard 7:2:1 split ratio. (a) Trend of test accuracy as the overlap rate increases. (b) Trend of the performance gap between training and test sets.

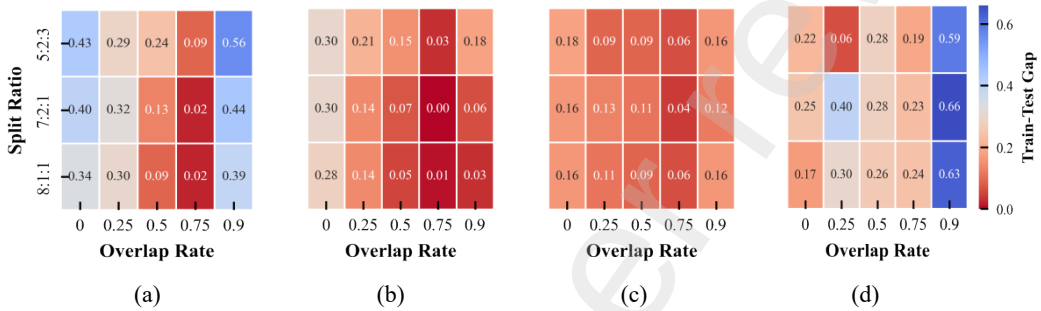
To mitigate random bias arising from parameter initialization and data sampling, all evaluation results are reported as the mean and standard deviation of five independent trials (with random seeds fixed at 42, 123, 456, 789, and 1010), utilizing 119,808 sampling points. The training set ratios are set to 50%, 70%, and 80%, which respectively correspond to three typical scenarios: a few-shot/underfitting environment, a standard academic benchmark, and a data saturation/overfitting limit. To comprehensively reveal the systematic impact of data leakage and eliminate the contingency associated with a single network architecture, a deep learning comparison matrix is constructed to cover four distinct feature extraction paradigms. The matrix includes a wide-kernel convolutional neural network (WDCNN) as the shallow local feature extraction benchmark, a 1D deep residual network (ResNet-1D) to explore the sensitivity of network depth to leaked data, a bidirectional long short-term memory network (C-BiLSTM) to reveal the shortcut learning tendency of temporal models under highly similar leaked samples, and a 1D Transformer to test whether the self-attention mechanism can resist underlying data distribution pollution at the algorithmic level. All models operate on a unified hardware and software platform featuring the Windows 11 operating system, Python 3.11, an AMD R9 9950X3D (4.40 GHz) processor, an NVIDIA 5090D v2 GPU, and 64 GB of RAM.

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4.1 Performance Analysis under Overlap Rate Induction

Sliding window sampling of continuous vibration signals inherently introduces strong temporal correlations between adjacent samples. When these highly redundant window samples are randomly assigned to training and test sets, the test set inevitably contains numerous waveform segments already exposed to the model during the training phase. This phenomenon violates the independent and identically distributed (i.i.d.) blind test assumption, subsequently leading to artificially inflated test accuracy. To quantitatively reveal this mechanism, three splitting strategies—random splitting, sequential splitting, and SDSP block isolation splitting—are systematically compared (as illustrated in Fig. 1). This comparison is conducted under the representative operating condition T3-s600 of the eccentric gear dataset, utilizing a baseline 7:2:1 split ratio across various overlap rates (0, 0.25, 0.50, 0.75, and 0.90).



(a) (b) (c) (d)

Fig. 4. Comparison of data leakage across different deep learning architectures and split ratios under the random splitting strategy for T3 s600: (a) WDCNN; (b) ResNet-1D; (c) C-BiLSTM; (d) Transformer.

Fig. 3 illustrates the trends in test accuracy and the train-test accuracy gap for ResNet-1D as the overlap rate varies. Under random splitting, the test accuracy exhibits a significant monotonic upward trend with increasing overlap. At zero overlap, the test accuracy of ResNet-1D is merely 0.70±0.05. However, as the overlap rate increases to 75%, the accuracy surges to 1.00±0.00, and the corresponding train-test accuracy gap plummets from 0.30±0.05 to 0.00±0.00, superficially presenting a nearly perfect generalization capability. This illusion, however, reverses drastically upon the introduction of temporal isolation. Under SDSP block isolation splitting, the test accuracy of ResNet-1D consistently hovers near the random guessing level (0.50±0.05 to 0.49±0.06) across all overlap rates. This stark contrast indicates that the ostensibly high accuracy largely benefits from cross-subset temporal redundancy rather than the genuine learning of physical fault features. Furthermore, the performance of sequential splitting closely aligns with that of block isolation splitting and diverges significantly from random splitting. This alignment demonstrates that performance inflation under random protocols primarily stems from the cross-subset temporal overlap introduced by global random shuffling, rather than from a mere increase in training data volume.

Table 1 summarizes the comprehensive numerical results for the 7:2:1 split ratio under condition T3-s600. Under random splitting, the test accuracy of WDCNN rises from 0.59±0.05 at zero overlap to 0.98±0.00 at 75% overlap, whereas under block isolation splitting, it consistently fluctuates between 0.38 and 0.54. C-BiLSTM and Transformer exhibit similar patterns; however, their absolute accuracies under block isolation splitting are notably lower. Specifically, the Transformer only reaches 0.35±0.03 under 75% overlap with block isolation and further deteriorates to 0.17±0.02 at 90% overlap, suggesting that attention-based models possess more fragile generalization capabilities when subjected to temporally isolated evaluation.

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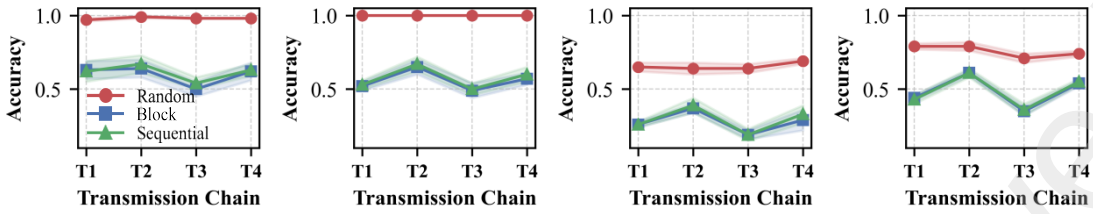


Fig. 5. Accuracy comparison across different transmission chains for S600: (a) WDCNN; (b) ResNet-1D; (c) C-BiLSTM; (d) Transformer.

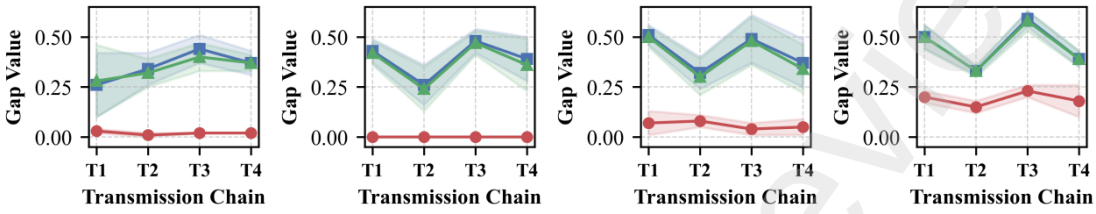


Fig. 6. Comparison of the performance gap (difference between training and test accuracy) across different transmission chains for S600: (a) WDCNN; (b) ResNet-1D; (c) C-BiLSTM; (d) Transformer.

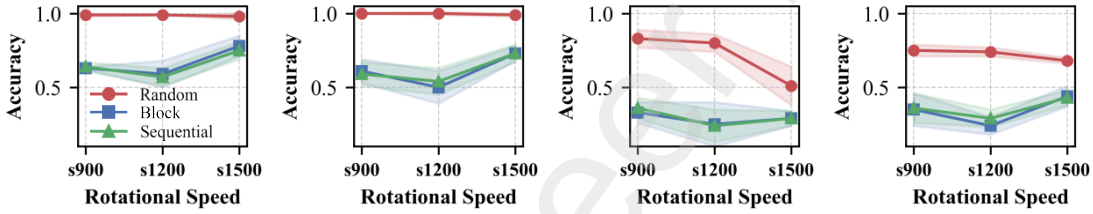


Fig. 7. Accuracy comparison under different rotational speeds for T3: (a) WDCNN; (b) ResNet-1D; (c) C-BiLSTM; (d) Transformer.

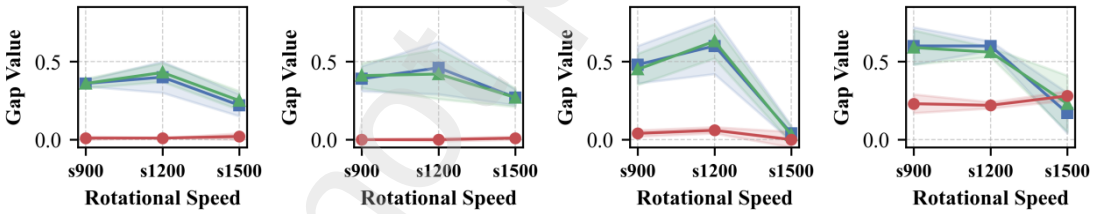


Fig. 8. Comparison of the performance gap (difference between training and test accuracy) under different rotational speeds for T3: (a) WDCNN; (b) ResNet-1D; (c) C-BiLSTM; (d) Transformer.

Notably, certain models (especially WDCNN and Transformer) exhibit a non-monotonic phenomenon where accuracy degrades at a 90% overlap rate under random splitting. While extremely high overlap rates exacerbate cross-subset redundancy, they simultaneously and severely compress the diversity of effective training samples. This compression causes the model to overfit local waveform segments, occasionally weakening its ability to recognize even the repetitive segments in the test set. Consequently, the leakage-induced accuracy peak typically emerges around a 75% overlap rate rather than under extreme overlap conditions. To further illustrate this, Fig. 4 displays a comparison heatmap of data leakage effects across different deep learning architectures and split ratios (5:2:3, 7:2:1, 8:1:1) under condition T3-s600 using the random splitting strategy. The results confirm that leakage-induced accuracy inflation is not an accidental phenomenon specific to a particular network architecture or split ratio. Instead, it manifests consistently across all compared conditions, thereby demonstrating significant systematicity.

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Figs. 5 and 6 present comparisons of the test accuracy and the train-test accuracy gap for the four models across different transmission chain configurations (T1–T4) at the S600 rotational speed. Under random splitting, the accuracy of all models from T1 to T4 remains generally high and tends to saturate. Conversely, under block isolation splitting, the accuracy exhibits greater volatility as the transmission chain lengthens, and the train-test accuracy gap widens significantly. This indicates the differential impact that various physical configurations have on leakage sensitivity. Furthermore, Figs. 7 and 8 show the trends in accuracy and the accuracy gap for three rotational speeds (s900, s1200, s1500) under the T3 configuration. As the rotational speed increases, the performance of all models under block isolation splitting improves slightly. This observation aligns with the underlying physical mechanism: stronger periodic fault features inherent in high-speed signals help reduce the model's reliance on local temporal redundancy.

Table 1. Accuracy and performance gap results for different models under the T3 s600 condition with a 7:2:1 split ratio.

Model	Splitting strategy	Metric	Overlap rate				
			0	0.25	0.5	0.75	0.9
WDCNN	Block	Acc	0.50 ± 0.11	0.53 ± 0.02	0.54 ± 0.07	0.50 ± 0.05	0.38 ± 0.02
		Acc Gap	0.43 ± 0.17	0.40 ± 0.08	0.42 ± 0.09	0.44 ± 0.07	0.61 ± 0.02
	Random	Acc	0.59 ± 0.05	0.68 ± 0.04	0.87 ± 0.04	**0.98 ± 0.00**	0.56 ± 0.03
		Acc Gap	0.40 ± 0.05	0.32 ± 0.04	0.13 ± 0.04	**0.02 ± 0.00**	0.44 ± 0.03
	Sequential	Acc	0.51 ± 0.10	0.53 ± 0.03	0.55 ± 0.05	0.54 ± 0.04	0.35 ± 0.02
		Acc Gap	0.42 ± 0.16	0.45 ± 0.06	0.41 ± 0.08	0.40 ± 0.07	0.64 ± 0.02
ResNet-1D	Block	Acc	0.50 ± 0.05	0.45 ± 0.07	0.50 ± 0.03	0.49 ± 0.05	0.49 ± 0.06
		Acc Gap	0.44 ± 0.11	0.51 ± 0.09	0.43 ± 0.08	0.48 ± 0.06	0.51 ± 0.06
	Random	Acc	0.70 ± 0.05	0.86 ± 0.04	0.93 ± 0.02	**1.00 ± 0.00**	0.94 ± 0.02
		Acc Gap	0.30 ± 0.05	0.14 ± 0.04	0.07 ± 0.02	**0.00 ± 0.00**	0.06 ± 0.02
	Sequential	Acc	0.50 ± 0.05	0.48 ± 0.04	0.50 ± 0.04	0.50 ± 0.04	0.49 ± 0.07
		Acc Gap	0.43 ± 0.11	0.47 ± 0.06	0.43 ± 0.08	0.47 ± 0.06	0.51 ± 0.07
C-BiLSTM	Block	Acc	0.17 ± 0.11	0.27 ± 0.10	0.26 ± 0.13	0.19 ± 0.03	0.33 ± 0.05
		Acc Gap	0.38 ± 0.08	0.27 ± 0.19	0.34 ± 0.21	0.49 ± 0.12	0.27 ± 0.27
	Random	Acc	0.50 ± 0.08	0.52 ± 0.04	0.57 ± 0.03	0.64 ± 0.03	0.46 ± 0.02
		Acc Gap	0.16 ± 0.07	0.13 ± 0.06	0.11 ± 0.05	0.04 ± 0.03	0.12 ± 0.05
	Sequential	Acc	0.20 ± 0.09	0.27 ± 0.10	0.26 ± 0.11	0.19 ± 0.04	0.31 ± 0.04
		Acc Gap	0.34 ± 0.11	0.28 ± 0.18	0.34 ± 0.19	0.48 ± 0.12	0.31 ± 0.28
Transformer	Block	Acc	0.27 ± 0.02	0.26 ± 0.04	0.31 ± 0.07	0.35 ± 0.03	0.17 ± 0.02
		Acc Gap	0.51 ± 0.15	0.55 ± 0.14	0.53 ± 0.06	0.59 ± 0.04	0.65 ± 0.30
	Random	Acc	0.57 ± 0.05	0.56 ± 0.06	0.64 ± 0.06	0.71 ± 0.03	0.32 ± 0.02
		Acc Gap	0.25 ± 0.12	0.40 ± 0.08	0.28 ± 0.13	0.23 ± 0.03	0.66 ± 0.02
	Sequential	Acc	0.27 ± 0.02	0.28 ± 0.04	0.31 ± 0.07	0.36 ± 0.04	0.17 ± 0.02
		Acc Gap	0.51 ± 0.15	0.58 ± 0.09	0.52 ± 0.04	0.58 ± 0.05	0.65 ± 0.28

In summary, overlap-induced performance inflation is a systematic evaluation bias that persists across various architectures, operating conditions, and split ratios. Its root cause lies in the fact that random splitting fundamentally violates the temporal independence assumption among continuous time series samples, rather than reflecting any true fault generalization capabilities possessed by the models.

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4.2 Evaluation of True Model Robustness Based on the SDSP Protocol

Building upon the qualitative revelation of the leakage phenomenon, this section introduces two quantitative metrics, namely the *LIF* and the Δ*Gap*, to systematically evaluate the true robustness of each model under different physical operating conditions. *LIF* measures the relative performance degradation of a model from random splitting to SDSP block isolation splitting, reflecting the absolute contribution of leakage to accuracy. Conversely, Δ*Gap* quantifies the extent to which traditional evaluation protocols mask the true train-test discrepancy, thereby revealing the deceptiveness of the evaluation results. The calculation formulas for these two metrics are presented in Equations (5) and (6). Here, the quantitative analysis utilizes the most mainstream 7:2:1 split ratio in the fault diagnosis field and a 75% overlap rate under random splitting as the baseline configuration, with the random seed fixed at 456.

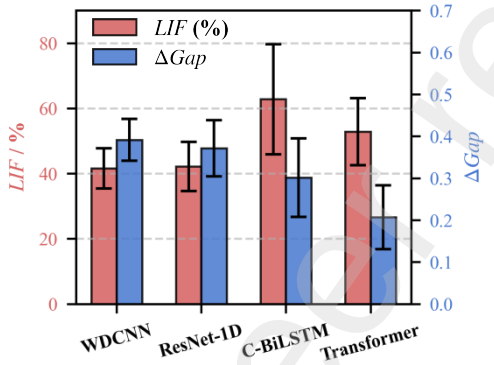


Fig. 9. Global statistical evaluation of the *LIF* and Δ*Gap* across different deep learning models.

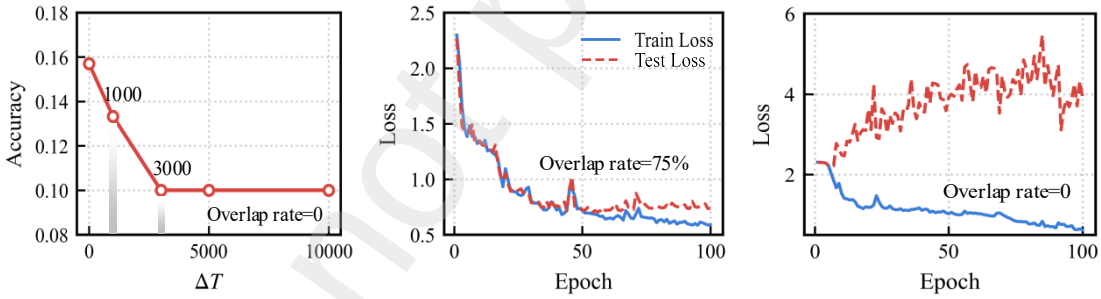


Fig. 10. Objective evaluation of the generalization ability of the C-BiLSTM model: (a) performance degradation curve vs. physical safety margin Δ*T*; (b) convergence behavior under random splitting; (c) convergence behavior under the SDSP protocol.

Fig. 9 summarizes the global statistical results of *LIF* and Δ*Gap* for the four models across seven physical operating conditions, while the detailed numerical values for each condition are provided in Table 2. Overall, all models exhibit measurable leakage vulnerability, although the differences among various architectures are quite significant. The global *LIF* values for WDCNN and ResNet-1D are 41.6±6.2% and 42.2±7.6%, respectively, with corresponding Δ*Gap* values of 0.39±0.05 and 0.37±0.07. This indicates that both convolutional and residual architectures can achieve high apparent accuracy under leakage conditions; however, once temporal isolation is enforced, their performance experiences substantial degradation.

Sequential and attention-based architectures demonstrate an even stronger dependence on leakage. The global *LIF* of C-BiLSTM reaches as high as 62.8±16.9%, with a Δ*Gap* of 0.30±0.09. Similarly, the Transformer yields a global *LIF* of 52.8±10.3% and a Δ*Gap* of 0.21±0.08. These

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results suggest that architectures equipped with long-range temporal modeling capabilities do not inherently possess leakage resistance. On the contrary, when temporal redundancy exists between data subsets, such models are more inclined to exploit local waveform similarities across subsets to construct shortcut representations, thereby presenting highly deceptive high accuracy under traditional evaluation protocols.

From the perspective of physical operating conditions, leakage sensitivity also exhibits obvious condition dependence. Under the T3-s600 condition, the *LIF* values for ResNet-1D and WDCNN reach 50% and 49%, respectively. Furthermore, C-BiLSTM exhibits extremely strong leakage vulnerability, with *LIF* values soaring to 75.4%, 76.6%, and 73.4% under the T1-s600, T2-s600, and T3-s600 conditions, respectively. As the rotational speed increases to 1500 rpm, the leakage sensitivity of most models decreases. Specifically, the *LIF* values for WDCNN, ResNet-1D, and C-BiLSTM drop to 29.6%, 29.3%, and 29.4%, respectively. This occurs because the more prominent periodic fault components in high-speed signals help enhance the model's ability to identify true fault features, thereby relatively reducing its reliance on local temporal redundancy.

Table 2. *LIF* and Δ*Gap* across different model architectures and physical operating conditions.

Model	Metric	T1-s600	T2-s600	T3-s600	T4-s600	T3-s900	T3-s1200	T3-s1500	Global
WDCNN	*LIF* (%)	40.2	39.4	49	44.9	43.4	44.4	29.6	41.6 ± 6.2
	Δ*Gap*	0.38	0.38	0.41	0.42	0.42	0.44	0.29	0.39 ± 0.05
ResNet-1D	*LIF* (%)	46	34	50	45	47	44	29.3	42.2 ± 7.6
	Δ*Gap*	0.4	0.29	0.44	0.33	0.44	0.41	0.29	0.37 ± 0.07
C-BiLSTM	*LIF* (%)	75.4	76.6	73.4	52.2	65.1	67.5	29.4	62.8 ± 16.9
	Δ*Gap*	0.46	0.35	0.34	0.25	0.3	0.24	0.17	0.30 ± 0.09
Transformer	*LIF* (%)	64.6	49.4	62	52.7	49.3	58.1	33.8	52.8 ± 10.3
	Δ*Gap*	0.27	0.12	0.28	0.22	0.2	0.27	0.09	0.21 ± 0.08

To further investigate the specific behavior of C-BiLSTM in fault diagnosis tasks, Fig. 10(a) displays the attenuation curve of test accuracy as the ΔT increases. As ΔT increases from 0 to approximately 3000 sampling points, the accuracy continuously declines from around 0.16 and eventually stabilizes at a low plateau. This indicates that the discriminative cues relied upon by the model possess strong temporal locality; once temporally adjacent information is isolated, the remaining features are insufficient to support robust fault classification. Figs. 10(b) and (c) compare the train-test loss convergence behaviors under random splitting (75% overlap) and the SDSP protocol, respectively. In the former, the training loss and test loss decrease synchronously and maintain high consistency, presenting a false impression of good convergence. In the latter, while the training loss continues to decrease, the test loss exhibits significant divergence or continuous oscillation, intuitively exposing the genuine generalization gap that exists under temporally isolated evaluation. This bifurcation characteristic in the loss curves is a typical dynamic hallmark of shortcut learning, indicating that the optimization objective of the model is not transferable fault features, but rather an overfitting to local waveform patterns within the training blocks.

In summary, leakage is not a binary attribute of a dataset, but rather a graded vulnerability jointly shaped by the sampling protocol, model architecture, and physical operating conditions. Through the joint evaluation of *LIF* and Δ*Gap*, the SDSP framework provides an actionable quantitative diagnostic tool for identifying whether model accuracy truly stems from physical

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generalization or merely from temporal memorization.

4.3 Feature Space Visualization and Shortcut Learning Mechanism Revelation

The significant performance difference between random splitting and SDSP indicates that overlapping windows fundamentally alter the internal representation structure learned by deep models. To deeply reveal this mechanism, WDCNN serves as a representative model under the T3-s600 condition. Quantitative evaluation is conducted through t-SNE dimensionality reduction visualization and the silhouette score.

Fig. 11 illustrates the feature space distributions under three different protocols. Under random splitting with a 75% overlap (Fig. 11(a)), the model forms compact and well-separated feature clusters. The silhouette score reaches as high as 0.53, and the four degradation states present clear boundaries within the feature space. However, this seemingly superior representation quality emerges under conditions where numerous temporally adjacent waveform segments are shared between the training and test sets. Its compactness largely reflects the repetition of local temporal patterns across subsets, rather than an essential distinction of physical fault categories.

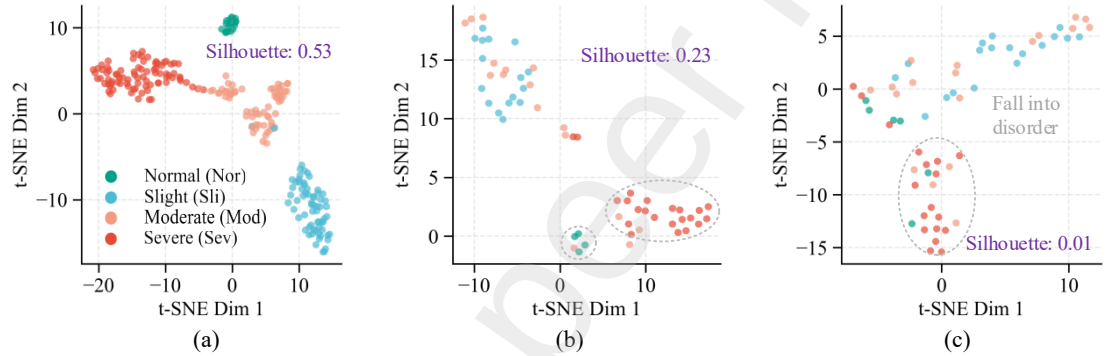


Fig. 11. Evolution of the feature space learned by the WDCNN model under different data splitting protocols: (a) random splitting with 75% overlap; (b) non-overlapping random splitting; (c) SDSP block-isolated splitting.

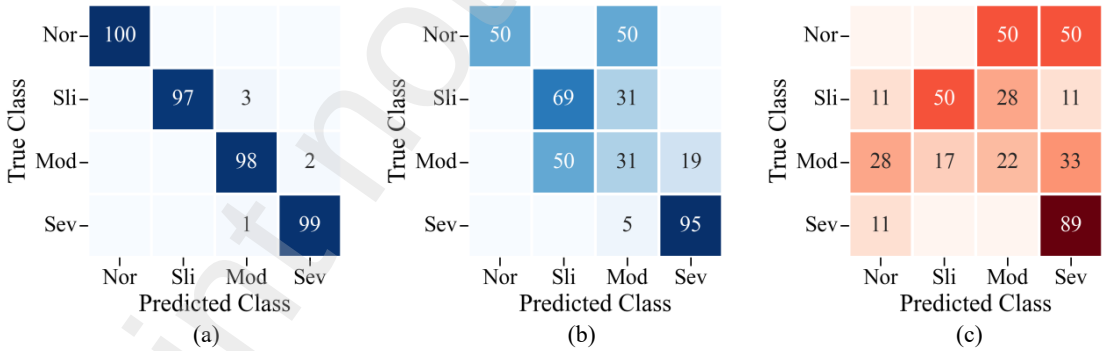


Fig. 12. Confusion matrices of the WDCNN model under different data splitting protocols: (a) random splitting with 75% overlap; (b) non-overlapping random splitting; (c) SDSP block-isolated splitting.

When the overlap rate drops to zero while retaining random splitting (Fig. 11(b)), the feature space structure exhibits obvious degradation. The silhouette score drops to 0.23, and the boundaries between adjacent degradation levels become blurred. Specifically, the confusion between the slight and moderate categories intensifies significantly. This result indicates that direct window overlap is the primary source of apparent representation quality. Nevertheless, non-overlapping random splitting still retains a certain degree of temporal proximity and operating condition similarity, which prevents the feature space from completely collapsing. Under SDSP block isolation splitting (Fig. 11(c)), the feature space falls into a highly disordered state. The silhouette score plummets to 0.01, and category boundaries become severely entangled. This

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demonstrates that once temporal leakage and boundary proximity are simultaneously eliminated, the representations learned under the traditional protocol lose their effective discriminative ability on temporally isolated test blocks.

The confusion matrices in Fig. 12 corroborate the aforementioned feature space analysis from the perspective of classification decisions. As shown in Fig. 12(a), under random splitting with a 75% overlap, the classification accuracy for all four fault types approaches a perfect score. Under zero-overlap random splitting (Fig. 12(b)), the confusion between the slight and moderate categories increases noticeably. While the recognition rate for the severe category remains at 95%, 50% of the samples in the normal category are misclassified. Under the SDSP protocol, as depicted in Fig. 12(c), classification performance experiences a precipitous drop. Only 50% of the samples in the slight category are correctly identified, and the accuracy for the moderate category plunges to 22%. Severe cross-confusion exists among all categories, indicating that the model essentially loses its ability to effectively discern the degree of degradation.

Overlapping random splitting enables the model to construct shortcut representations based on repetitive waveform segments and local temporal continuity. Such representations perform excellently under traditional evaluation protocols but fail to transfer to temporally isolated, unseen data blocks. By removing these shortcuts, SDSP effectively verifies whether the model truly learns physically meaningful and transferable fault features.

4.4 Validation Based on the CWRU Dataset

To verify whether the aforementioned leakage effect is confined to a specific experimental platform or possesses broader universality, the same analytical procedure is repeated on the Case Western Reserve University (CWRU) bearing fault benchmark dataset. The CWRU dataset serves widely as a benchmark in the fault diagnosis field. Numerous studies report near-perfect test accuracies under random splitting protocols on this dataset. This characteristic makes it an ideal external dataset for verifying whether high accuracy inherently relies on the evaluation protocol. Table 3 summarizes the test accuracy and accuracy gap results of the four models under different data splitting strategies. Under random splitting with a 75% overlap, all models achieve extremely high test accuracies. Specifically, WDCNN reaches 100.00±0.00%, ResNet-1D reaches 99.91±0.17%, C-BiLSTM reaches 99.74±0.21%, and Transformer reaches 98.79±1.00%. The corresponding accuracy gaps all approach zero, superficially presenting a nearly perfect generalization capability.

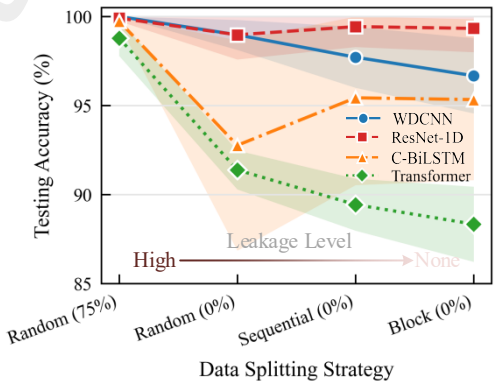


Fig. 13. Accuracy variations of different models under various data splitting strategies on the CWRU dataset.

However, as the degree of leakage systematically decreases, model performance exhibits a consistent decline. Under block isolation splitting with zero overlap, the accuracy of WDCNN

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drops to 96.67±2.11%, C-BiLSTM drops to 95.33±4.52%, and Transformer drops to 88.33±2.11%. The train-test accuracy gaps widen to 3.18±2.01, 2.02±1.92, and 11.67±2.11, respectively. ResNet-1D demonstrates relatively strong stability, maintaining an accuracy of 99.33±1.33% under block isolation splitting. This indicates that this model can extract more physically separable fault features in the CWRU task. Fig. 13 intuitively presents the accuracy trends of the four models as the data splitting strategy changes. The monotonic decrease in leakage level, from high (random 75%) to none (block isolation 0%), corresponds to a systematic decline in the accuracy of each model. Among them, the Transformer exhibits the most prominent performance drop.

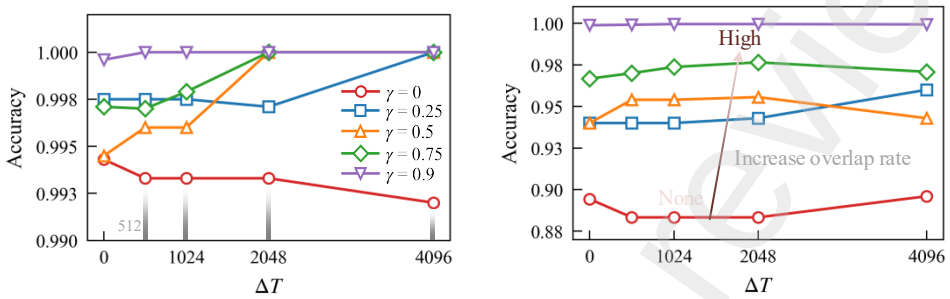


Fig. 14. Accuracy variations under varying overlap rates and Δ*T* on the CWRU dataset: (a) ResNet-1D; (b) Transformer.

The *LIF* and Δ*Gap* results reported in Table 4 further quantify the leakage sensitivity on the CWRU dataset. The Transformer yields the highest *LIF* at 10.58%, with a Δ*Gap* of 0.10. C-BiLSTM follows, with an *LIF* of 4.42% and a Δ*Gap* of 0.02. WDCNN has an *LIF* of 3.33% and a Δ*Gap* of 0.03. ResNet-1D shows the lowest *LIF* at merely 0.58%, with a Δ*Gap* of 0.01. Compared to the eccentric gear dataset, the overall *LIF* and Δ*Gap* values on the CWRU dataset are smaller. This reflects the stronger category separability and lower condition complexity in the CWRU task. Nevertheless, the directional changes of these metrics remain highly consistent with those observed in the eccentric gear dataset. This confirms that the systematic overestimation of model performance by traditional random splitting is not a unique phenomenon confined to a single experimental platform.

Table 3. Accuracy and performance gap results on the CWRU dataset under different data splitting strategies.

Model	Metric	Splitting strategy			
		Random (75%)	Random (0%)	Sequential (0%)	Block (0%)
WDCNN	Acc	100.00 ± 0.00	98.97 ± 0.84	97.71 ± 1.71	96.67 ± 2.11
	Acc Gap	0.00 ± 0.00	1.03 ± 0.84	2.29 ± 1.71	3.18 ± 2.01
ResNet-1D	Acc	99.91 ± 0.17	98.97 ± 1.38	99.43 ± 1.14	99.33 ± 1.33
	Acc Gap	0.06 ± 0.16	0.49 ± 1.18	0.57 ± 1.14	0.67 ± 1.33
C-BiLSTM	Acc	99.74 ± 0.21	92.76 ± 5.91	95.43 ± 4.90	95.33 ± 4.52
	Acc Gap	0.00 ± 0.20	4.29 ± 3.55	2.32 ± 2.22	2.02 ± 1.92
Transforme	Acc	98.79 ± 1.00	91.38 ± 1.09	89.43 ± 1.46	88.33 ± 2.11
r	Acc Gap	1.20 ± 0.99	8.62 ± 1.09	10.57 ± 1.46	11.67 ± 2.11

Table 4. *LIF* and Δ*Gap* for different models on the CWRU dataset.

Metric	Model			
	WDCNN	ResNet-1D	C-BiLSTM	Transformer
LIF (%)	3.33%	0.58%	4.42%	10.58%
Δ*Gap*	0.03	0.01	0.02	0.10

Fig. 14 illustrates the accuracy variations of ResNet-1D and Transformer under different combinations of overlap rates and Δ*T*. ResNet-1D maintains near-saturation accuracy under most overlap rate and Δ*T* settings, reflecting its strong true generalization capability in the CWRU task.

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In contrast, the Transformer exhibits a clear dependence on the overlap rate. Under zero-overlap conditions, its accuracy ranges from approximately 0.89 to 0.90. However, as the overlap rate increases to 90%, the accuracy approaches 1.00. This pattern remains consistent across different ΔT settings. It indicates that even on a relatively simple dataset, self-attention models can still achieve significant accuracy gains through overlap-induced redundancy. The aforementioned cross-dataset validation results demonstrate that leakage-induced evaluation bias is not an isolated artifact of a specific experimental platform. Instead, it is a prevalent systematic flaw within standard evaluation procedures for continuous vibration data. The magnitude of the bias depends on the fault separability and condition complexity of the dataset. However, its underlying mechanism remains consistent. Random splitting after windowing continuous time series allows temporally redundant information to infiltrate the test set. This systematically overestimates the true generalization capability of the models. Additional experimental results are provided in the supplementary materials.

5. Discussion

5.1 Shortcut Learning Mechanisms and Architectural Vulnerability

The experimental results demonstrate that the artificially inflated test accuracy induced by random overlapping splitting is systematic across architectures, operating conditions, and split ratios. The core significance of this finding does not lie in negating the architectural designs of existing models, but rather in highlighting a widely overlooked methodological risk within the evaluation procedures for continuous time series. The performance metrics measured under traditional protocols largely reflect the model's memory retrieval capability for cross-subset temporal redundancy, rather than its true generalization capability regarding physical signal features. The comparative results of t-SNE visualizations and confusion matrices provide direct evidence at the representation level: the same model forms compact and separable feature clusters (with a silhouette score of 0.53) under random splitting, whereas the feature space falls into severe disorder (silhouette score of 0.01) and classification performance experiences a precipitous drop upon switching to the SDSP protocol.

From an architectural perspective, leakage vulnerability exhibits a positive correlation trend with temporal modeling capability, a phenomenon that contradicts intuitive expectations. The global *LIF* values of C-BiLSTM and Transformer (62.8% and 52.8%, respectively) are significantly higher than those of convolutional architectures (approximately 41–42%). This indicates that while gating and self-attention mechanisms capture long-range temporal dependencies, they also render the models more susceptible to utilizing temporal continuity between training and test blocks as a discriminative cue for classification. Consequently, these models exhibit a stronger tendency toward shortcut learning under leakage conditions. Physical operating conditions also modulate leakage sensitivity. When the rotational speed increases to 1500 rpm, the *LIF* of each model decreases but still remains around 29%. This demonstrates that even under operating conditions with strong signal separability, the impact of the chosen evaluation protocol on the final results cannot be ignored.

Beyond specific architectures, the validation results on the CWRU dataset further reveal the interactive relationship between the leakage effect and the inherent separability of the dataset. Although the overall *LIF* of this dataset is relatively small, the Transformer still reaches an *LIF* of 10.58%, and its accuracy exhibits a clear monotonic upward trend as the overlap rate increases.

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This provides a more cautious perspective for interpreting the numerous near-perfect results reported on the CWRU dataset in existing literature. High accuracy alone is insufficient to prove that a model has truly learned transferable fault representations; deep temporal overfitting may be hidden beneath such results.

5.2 Methodological Extension of the SDSP Framework

Methodologically, the design logic of the SDSP framework is not confined to rotating machinery fault diagnosis but can be generalized to all learning tasks based on sliding window sampling of continuous time series. Whether dealing with biomedical ECG/EEG signals, climate and atmospheric records, financial time series, or industrial process monitoring data, the issue of cross-subset temporal redundancy introduced by random splitting objectively exists as long as samples are generated by slicing continuous signals. By enforcing a zero-overlap constraint and introducing the safety margin ΔT between the training and test sets, SDSP explicitly separates data augmentation from evaluation leakage at the operational level. Overlapping windows can still be utilized to expand training sample diversity, provided no redundancy is created across subsets. The physical significance of ΔT is equally universal: it corresponds to the temporal correlation length determined by system inertia, low-frequency background dynamics, or state persistence within specific signals. Its reasonable value can be estimated in a data-driven manner, similar to the accuracy attenuation curve of C-BiLSTM. As supplementary evaluation metrics, *LIF* and ΔGap can retrospectively diagnose the reliability of existing results without altering the underlying model architecture.

5.3 Implications for Transfer Learning and Industry Standardization

The findings of this study also hold significant implications for transfer learning and domain adaptation research. Currently, numerous domain adaptation studies attribute the performance degradation observed between source and target domains to domain shift, thereby driving the design of alignment algorithms. However, the experimental evidence presented herein suggests that if the source domain model itself is trained under temporal leakage conditions, the extracted representations may largely be memory encodings of local waveform redundancy in the source domain, rather than physical fault features possessing true cross-domain universality. Such memory-based representations perform excellently under random splitting evaluation in the source domain but fail rapidly upon domain transfer. This implies that the widely observed “insufficient transfer performance” problem in transfer learning may partially stem not merely from domain shift, but from the source domain model’s failure to genuinely learn transferable physical features during the initial stage. From this perspective, source domain pre-training established upon a leakage-free evaluation protocol is not only a prerequisite for ensuring evaluation credibility but also a critical foundational condition for driving substantial breakthroughs in transfer learning for fault diagnosis. Ultimately, subsequent domain alignment and feature transfer hold substantive meaning only when the model has truly anchored physical fault features in the source domain.

Fundamentally, the objective of this research is not to diminish the value of existing fault diagnosis methods, but rather to establish fairer and more rigorous evaluation standards, thereby providing a reliable methodological foundation for model comparison and performance reporting in this field. Currently, the diverse and inconsistent data splitting methods in the fault diagnosis field result in a lack of a unified benchmark for performance claims across different studies. By introducing the safety margin ΔT and temporal isolation mechanisms, the SDSP framework holds

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the promise of breaking this status quo and providing a practical solution for unifying industry evaluation standards. This will not only make horizontal comparisons between future algorithms more objective and transparent but also effectively bridge the gap between laboratory test metrics and true performance in industrial settings. Moving forward, it is recommended to adopt SDSP as the standard splitting protocol in the evaluation practices of fault diagnosis models, and to simultaneously introduce *LIF* and ΔGap as leakage-aware supplementary metrics alongside traditional performance indicators such as accuracy and recall. Only by establishing a unified and scientific evaluation benchmark can the actual capabilities of models under unknown operating conditions be comprehensively characterized, thereby truly propelling deep learning-based fault diagnosis technologies from theoretical research to large-scale industrial application.

6. Conclusion

The reliable evaluation of fault diagnosis models is a critical prerequisite for promoting the practical application of intelligent operation and maintenance. However, the random overlapping splitting strategy commonly adopted in existing studies inevitably introduces cross-subset temporal redundancy in continuous time series, leading to systematically inflated test accuracy. This study proposes the Standardized Data Splitting Protocol (SDSP) to fundamentally eliminate evaluation leakage through the joint constraints of temporal isolation and a safety margin (ΔT). Furthermore, the proposed framework is systematically validated across multi-architecture and multi-condition scenarios. The main conclusions are drawn as follows:

1. The inflationary effect of random overlapping splitting on test accuracy exhibits significant systematicity across various architectures and operating conditions. This evaluation bias is rooted in the data splitting protocol itself, rather than stemming from the conventional overfitting behavior of specific models.
2. Architectures equipped with long-range temporal modeling capabilities (e.g., C-BiLSTM and Transformer) exhibit stronger leakage vulnerability, with global *LIF* values reaching 62.8% and 52.8%, respectively. This indicates that they are more prone to falling into shortcut learning under leakage conditions.
3. Feature space analysis confirms the shortcut learning mechanism at the representation level. The feature silhouette score is 0.53 under random splitting but plummets to 0.01 after switching to SDSP. This demonstrates that models evaluated under traditional splitting tend to overfit local waveform redundancy rather than extract truly transferable physical fault features.
4. Cross-dataset validation on the CWRU dataset reveals that leakage-induced evaluation bias does not depend on a specific experimental platform. High accuracy under traditional evaluation is inherently insufficient to prove that a model possesses genuine detection and generalization capabilities.
5. The SDSP framework possesses broad universality and is applicable to all learning tasks based on sliding window sampling of continuous time series. As leakage-aware supplementary metrics, *LIF* and ΔGap are recommended to be incorporated into the standard evaluation system for fault diagnosis models alongside traditional performance indicators, thereby facilitating the unification of industry evaluation benchmarks.

Future work will advance along two primary directions. First, the reasonable value selection of the ΔT will be systematically investigated to establish universal ΔT selection criteria based on

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signal autocorrelation structures and system dynamic characteristics. Second, the general magnitude range of *LIF* will be examined across a broader spectrum of continuous time series tasks. This will provide reference benchmarks for leakage risk assessment in different application scenarios, thereby further promoting the standardized popularization and industrial application of the SDSP framework.

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